Primer: Adversarial Inverse Reinforcement Learning (AIRL)

Fu, Luo, and Levine (2018)

econirl package documentation

1 Context and Problem

Maximum entropy IRL [Ziebart, 2010] recovers a reward function that rationalizes observed behavior, but the recovered reward entangles the true reward with the value function. Formally, for any bounded function $h : \mathcal{S} \to \mathbb{R}$, the shaped reward

$$r'(s, a) \;=\; r(s, a) \;+\; \gamma\, \mathbb{E}_{s' \sim P(\cdot|s,a)}[h(s')] \;-\; h(s) \tag{1}$$

produces the same optimal policy under the original transitions P [Ng et al., 1999]. If P changes (new environment, counterfactual), the shaping term evaluates differently and the shaped reward no longer rationalizes the same behavior. This matters in economics because counterfactual policy evaluation requires a reward that transfers across environments.

Fu et al. [2018] propose AIRL to disentangle the transferable reward from the non-transferable value function. The key idea: structure the discriminator so that one component converges to the true reward $r^*(s)$ and the other absorbs the value function $V^*(s)$. At the adversarial optimum, the reward component is invariant to changes in dynamics.

2 Model

2.1 Notation

Symbol	Meaning
$s \in \mathcal{S}$, $a \in \mathcal{A}$, s'	State, action, next state
$\gamma \in [0, 1)$	Discount factor
$P(s' \mid s, a)$	Transition kernel
$r^*(s)$	True reward (state-only; Fu et al. 2018 require this for Theorem 1)
$\pi^*(a \mid s)$	Expert policy (observed through demonstrations)
$\tau = (s_0, a_0, s_1, a_1, \ldots)$	Trajectory
$V^*(s)$, $Q^*(s, a)$	Value function, action-value function
$A^*(s, a) = Q^*(s, a) - V^*(s)$	Advantage function
$g_\theta(s)$	Reward approximator (param. θ); state-only
$h_\phi(s)$	Shaping potential $\approx V(s)$ (param. ϕ)
$\pi_\omega(a \mid s)$	Learned policy (param. ω)
$f_{\theta,\phi}(s, a, s')$	Discriminator logit (4)
$D_{\theta,\phi}(s, a, s')$	Discriminator output $\in [0, 1]$ (5)

2.2 Maximum Causal Entropy Objective

The agent maximizes entropy-regularized cumulative reward [Ziebart, 2010]:

$$\max_{\pi} \ \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t \big(r(s_t, a_t) + \mathcal{H}(\pi(\cdot \mid s_t))\big)\right]. \tag{2}$$

The optimal policy is the softmax over advantages:

$$\pi^*(a \mid s) = \exp\big(A^*(s,a)\big) \Big/ \textstyle\sum_{a'} \exp\big(A^*(s,a')\big). \tag{3}$$

Taking the log: $\log \pi^*(a \mid s) = A^*(s,a) = Q^*(s,a) - V^*(s)$.

2.3 Discriminator Structure

AIRL uses a structured discriminator that decomposes into reward and shaping:

$$f_{\theta,\phi}(s,a,s') = g_\theta(s) + \gamma\, h_\phi(s') - h_\phi(s), \tag{4}$$

$$D_{\theta,\phi}(s,a,s') = \frac{\exp(f_{\theta,\phi})}{\exp(f_{\theta,\phi}) + \pi_\omega(a \mid s)}. \tag{5}$$

The logit f mirrors the Bellman relationship $Q(s,a) = r(s) + \gamma \mathbb{E}[V(s')]$, decomposed into $g_\theta \approx r$ and $h_\phi \approx V$. Note that g_θ depends on s only, not on a. The discriminator classifies transitions as expert ($D \to 1$) or generated ($D \to 0$).

3 Identification and Theorems

Reward ambiguity (background). IRL rewards are identified only up to potential-based shaping (1) [Ng et al., 1999]. Cao and Cohen [2021] formalize this: the observed policy constrains $|\mathcal{S}|(|\mathcal{A}| - 1)$ values but the reward has $|\mathcal{S}| \cdot |\mathcal{A}|$ unknowns, leaving $|\mathcal{S}|$ free variables corresponding to the value function.

Decomposability (Fu et al., Definition B.1). An MDP is *decomposable* if all states are linked: for every pair (s, s') there exists a policy and a time t such that s' is reachable from s with positive probability. This is a standard ergodicity-like condition that rules out absorbing states.

Disentanglement (Fu et al., Theorem 1). If (a) g_θ is state-only, (b) the MDP is decomposable, and (c) the adversarial training converges to the Nash equilibrium, then $g_\theta(s) = r^*(s) + c$ for a constant c.

The proof uses the fact that at the Nash equilibrium, $D = 1/2$ everywhere, so $f_{\theta,\phi} = \log \pi^* = A^*$. Substituting the decomposition: $g_\theta(s) + \gamma h_\phi(s') - h_\phi(s) = r^*(s) + \gamma V^*(s') - V^*(s)$. Since g_θ depends on s only, taking the expectation over s' gives $g_\theta(s) = r^*(s) + h_\phi(s) - V^*(s) + c_0$. The decomposability condition then implies $h_\phi(s) - V^*(s)$ is constant across all states, so $g_\theta(s) = r^*(s) + c$.

Limitations. Theorem 1 requires state-only rewards. If the true reward depends on actions ($r^*(s,a)$), as in DDC models where different actions have different payoffs, the disentanglement guarantee breaks. Lee et al. [2026] resolve this with anchor constraints (see AAIRL primer). The decomposability condition excludes absorbing states, which appear in every DDC model with an exit option.

4 Estimation Algorithm

Algorithm 1 AIRL (Fu et al. 2018)

Require: Expert demonstrations $\mathcal{D}_E = \{\tau_1, \ldots, \tau_N\}$, discount γ
Ensure: Disentangled reward $g_\theta(s)$

```
Initialize θ_0, φ_0, ω_0
for iteration = 1, 2, ... do
    // Step 1: Sample from current policy
    D_π ← roll out π_ω in environment to collect M transitions
    // Step 2: Train discriminator
    Extract expert transitions: {(s, a, s')} ~ D_E
    Extract policy transitions: {(s, a, s')} ~ D_π
    for j = 1, ..., J discriminator steps do
        f_n ← g_θ(s_n) + γ h_φ(s'_n) − h_φ(s_n)                    // Eq. (4)
        D_n ← σ(f_n − log π_ω(a_n | s_n))                          // Eq. (5), σ = sigmoid
        ← −E_{D_E}[log D] − E_{D_π}[log(1 − D)]
        θ, φ ← θ, φ − α_D ∇_{θ,φ}
    end for
    // Step 3: Update policy
    Compute reward signal for policy: r̃(s, a, s') ← log D − log(1 − D) = f − log π_ω
    Update ω via any policy optimization method (PPO in practice)
    // Step 4: Check convergence
    if D ≈ 1/2 on held-out data then
        break
    end if
end for
return g_θ    (disentangled reward, transferable across dynamics)
```

Key implementation details. The reward signal in Step 3 is $\tilde{r} = \log D - \log(1 - D)$, which equals $f - \log \pi_\omega$ by construction of (5). This is the log-odds of the discriminator and serves as the advantage estimate. Fu et al. [2018] use PPO [Schulman et al., 2017] for the policy update (Step 3), though any policy optimization method works. The discriminator and policy are trained alternately, with $J = 1$ discriminator step per policy step in their experiments.

Convergence. At the Nash equilibrium, $D = 1/2$ everywhere and $f_{\theta,\phi} = A^*$. By Theorem 1, $g_\theta(s) = r^*(s) + c$. The constant c is economically irrelevant because it does not affect any policy or counterfactual prediction.

5 Results from the Paper

Fu et al. [2018] test AIRL on continuous-control tasks from OpenAI Gym and transfer experiments.

Transfer experiment. They train AIRL on a PointMaze environment with one set of walls, then evaluate the recovered reward in a maze with different walls. AIRL transfers successfully (the agent navigates the new maze), while GAIL and MaxEnt IRL fail because their recovered rewards are

entangled with the original dynamics. This is the paper's central result: disentanglement enables transfer.

Continuous control. On Ant, HalfCheetah, and Hopper, AIRL matches or exceeds GAIL in terms of policy performance (cumulative reward) while also recovering a reward function. GAIL recovers a policy but not a reward.

State-only vs. state-action. When the true reward is state-only, AIRL with $g_\theta(s)$ recovers it accurately (Theorem 1). When the true reward depends on actions, $g_\theta(s)$ absorbs the action-dependent variation into the shaping potential, and the recovered reward is less interpretable.

6 Simulation

We test AIRL on a 5×5 gridworld with state-only reward (goal at corner), 4 actions (N/S/E/W), stochastic transitions (80% intended, 20% random), and $\gamma = 0.95$. The true reward is $r^*(s) = 1$ at the goal and $r^*(s) = 0$ elsewhere. Expert demonstrations come from the optimal soft Bellman policy. Run `python airl_run.py` then recompile to update Table 1.

Metric	True	AIRL
Cosine similarity	– – –	0.000
Correlation	– – –	0.000
Policy match rate	– – –	0.000
Goal reward	1.00	0.000
Mean non-goal reward	0.00	0.000

Table 1: AIRL reward recovery on 25-state gridworld with state-only reward. Auto-generated by `airl_run.py`.

The simulation uses the tabular AIRL estimator with conservative policy iteration ($\alpha_p = 0.1$) to stabilize the adversarial loop. On this small MDP where the decomposability condition holds (all states are reachable), AIRL should recover the goal reward up to a constant. The recovered reward is compared to the true reward via cosine similarity and policy match rate (fraction of states where the greedy action agrees).

7 Comparison

	MaxEnt IRL	MCE-IRL	GAIL	AIRL	AAIRL
Recovers reward	Yes	Yes	No	Yes	Yes
Reward transfers	No	No	N/A	Yes	Yes
Action-dep. r	Yes	Yes	N/A	No*	Yes
Continuous states	No	No	Yes	Yes	Yes
Requires P	Yes	Yes	No	No	No†

Table 2: *Theorem 1 requires state-only $g_\theta(s)$. †With PPO generator; tabular variant uses known P.

AIRL's contribution over MaxEnt/MCE-IRL is reward transferability. Its contribution over GAIL is that it recovers a reward function, not just a policy. Its limitation relative to AAIRL [Lee

et al., 2026] is the state-only restriction: Theorem 1 does not apply to action-dependent rewards, which are the norm in DDC models. AAIRL resolves this with anchor constraints (see AAIRL primer).

References

H. Cao and S. N. Cohen. Identifiability in inverse reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

J. Fu, K. Luo, and S. Levine. Learning robust rewards with adversarial inverse reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2018.

P. Lee, K. Sudhir, and Y. Wang. Modeling serialized content consumption: Adversarial IRL for dynamic discrete choice. Working paper, 2026.

A. Y. Ng, D. Harada, and S. Russell. Policy invariance under reward transformations: Theory and application to reward shaping. In *International Conference on Machine Learning (ICML)*, 1999.

J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. In *arXiv preprint arXiv:1707.06347*, 2017.

B. D. Ziebart. Modeling purposeful adaptive behavior with the principle of maximum causal entropy. In *Ph.D. Thesis, Carnegie Mellon University*, 2010.